Seeing with Our Hands, Hearing with Our Eyes: Accessible AI in Samsung Displays for Inclusion and Sustainability

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In 2025, more than 2.2 billion people—nearly one in four worldwide—will live with some form of visual impairment [1], and about 430 million (5% of the world's population) will experience disabling hearing loss [2], according to the World Health Organization (WHO). In Mexico, hearing loss amounts to 19,376 cases per 100,000 inhabitants, a figure substantially higher than in developed countries such as the United States or Canada [3].

These figures demand a redesign of technological tools and devices, transforming them into affordable solutions that address the full spectrum of needs: from limitations in mobility, sight, and hearing to differences in intellectual functioning.

This essay examines how Samsung Electronics, a global player in electronics, semiconductors, mobile phones, and televisions (with a 28.3% share of the worldwide television market in 2024 [4]), has evolved in integrating accessibility features into its smart displays.

I argue that AI-powered accessibility innovation is not merely a way to meet corporate ethical responsibilities: it is a social component aligned with the United Nations Sustainable Development Goals (SDGs). Moreover, empirical evidence shows that disability inclusion yields tangible competitive advantages. An Accenture study examining 346 organizations found that disability-inclusion leaders generated 1.6 times the revenue and 2.6 times the net income of their peers [5]. These figures indicate that commitment to accessibility can also generate innovation and economic value.

Inaccessible technology undermines independence and hinders work and learning. The data suggest that 44% of website users would change companies if a technology were not accessible; 79% report insufficient independence, and 63% say they have few choices among businesses offering inclusive technological solutions [6]. The same problem appears in audiovisual devices: complicated menus, remote controls without tactile or audio feedback, and content without alternative descriptions produce exclusionary experiences.

The United Nations Convention on the Rights of Persons with Disabilities recognizes access to information and communication technology as an essential human right [7]. In this context, Samsung's accessible displays relate to several SDGs: SDG 10 (Reduced Inequalities) by broadening access to audiovisual content; SDG 4 (Quality Education) by facilitating educational content for people with disabilities; SDG 8 (Decent Work and Economic Growth) by supporting independence needed for employment; and SDG 12 (Responsible Consumption and Production) through durable, useful products [8].

Environmental and social sustainability are also intertwined. Accessible, well-designed products can last longer because people with disabilities may retain a tool that adapts to their changing needs. This principle underlies the circular economy: prioritizing durability, adaptability, and longevity over frequent replacement reduces electronic waste, makes better use of resources, and limits demand for virgin raw materials. In this sense, accessibility can be a catalyst for environmental sustainability.

Samsung's innovation ecosystem

To put these commitments into practice, Samsung has developed an ecosystem of accessible innovations. Among its most significant is Relumino Mode, introduced at CES 2023 and recognized with a CES Innovation Award. The feature is designed for people with low vision rather than complete blindness, a group described in the original source as accounting for 90% of people with visual impairment. Relumino sharpens contours and adjusts contrast, colors, and definition so viewers can distinguish objects on the screen, improving an otherwise blurred or inaccessible audiovisual experience [9].

Relumino Mode underwent clinical testing at Samsung Medical Center. Participants with low vision viewed televisions from one meter away under controlled lighting conditions. The tests included subjective measures (user feedback and satisfaction) and objective measures (contrast sensitivity assessments using scientific instruments) [10]. Relumino also received the Royal National Institute of Blind People's “Tried and Tested” accreditation in the

United Kingdom, with Samsung becoming the first television manufacturer to receive it [11]. David Clark of RNIB said that, for the first time, people who are blind or partially sighted could use a smart television fully and independently [12].

The development process also followed the principle of designing with users, not just for them. Samsung engineers worked with visually impaired people and developed special simulation glasses to understand the varied and complex experiences of reduced vision (Samsung Electronics, 2023). The original essay reports that this methodology can improve usability by 40% [13]. More importantly, it illustrates design with agency and social responsibility: users are not passive consumers, but co-creators of technologies that shape their futures.

In 2023 Samsung also introduced SeeColors Mode for users with color-vision deficiency. It adjusts red, green, and blue levels for different types of color-vision impairment and received TÜV Rheinland certification for color-vision accessibility [14].

Samsung further offers a multimodal range of accessibility solutions [15].

For people who are blind or have low vision:

• Voice Guide: announces menu options, volume changes, channels, and program information, with customizable tone and speaking speed.

• Audio Description: provides spoken descriptions of content on screen, allowing blind users to follow visual narratives.

• High Contrast: displays white text on an opaque black background in menus for users with low vision.

• Zoom Menu & Text: enlarges text and interface elements.

• Grayscale Mode: displays only black, white, and gray, potentially helping users with severe color-vision difficulties.

• Color Inversion: reverses interface colors to produce gentler visual experiences for some users.

• Learn TV Remote: verbally explains the functions of remote-control buttons and screen elements.

• Picture Off: switches off the image while retaining audio, useful for blind viewers.

For people who are deaf or hard of hearing:

• Closed Captions: offer settings for text size, font, color, and background.

• Sign Language Zoom: enlarges and repositions an interpreter on screen.

• Sign Language Guide: uses an avatar to interpret menu options in American Sign Language (ASL), with customizable size, speed, and placement.

• Multi-Output Audio: plays audio through both television speakers and Bluetooth headphones, supporting viewers who use hearing devices.

On September 4, 2025, at IFA in Berlin, Samsung announced Bixby-powered Vision AI for personalized conversational experiences [16]. The ecosystem includes [17]:

• Live Translate: real-time translation of conversations and on-screen dialogue, with the potential to reduce language barriers for deaf and hard-of-hearing viewers when subtitles are unavailable in their language.

• Voice-based Scene Description: allows a Vision AI companion to describe on-screen elements on request, potentially extending independence beyond prerecorded audio descriptions.

• Active Voice Amplifier Pro: adapts voice clarity to surrounding noise.

• AI Picture Optimization: adjusts contrast and brightness without requiring extensive manual configuration, potentially benefiting people with low vision.

AI thus makes it possible for displays not only to show content but also to learn, decide, and adapt dynamically to an individual's needs.

The next challenge: intersecting disabilities

Despite these advances, major challenges remain, especially when disabilities intersect. What happens when a person is both deaf and blind? Television accessibility for deafblind people depends on combining haptic technology, braille, and personalized software; these solutions have not yet reached the accessibility levels available for many other sensory disabilities.

Some possible approaches are being explored in research on electronic braille devices. Current devices can incorporate vibration patterns to convey the events or emotions described in audiovisual narratives [18]. In sports, 5G-connected tablets already translate plays and actions into tactile vibrations and movements for blind people; in theory, these could be adapted for deafblind audiences if supplemented with braille channels [19].

Other possible approaches include voice-to-braille software, which converts narration and subtitles into tactile braille [20], and emerging brain–computer interfaces, including the use of neural earbuds to explore device control and the communication of audiovisual events through electroencephalographic signals [21].

Although Samsung already offers a sign-language avatar for menu options, it has not been extended throughout broadcast channels and television series. In 2021 a system for recognizing sign language through a camera integrated into televisions was proposed (CNET, 2021). Technical obstacles remain substantial: real-time computer vision, AI models trained in different sign languages, and the diversity of more than 300 regional sign languages with significant variation [22].

According to the US Centers for Disease Control and Prevention, cognitive disability is the second most prevalent category of disability [23]. Future designs should therefore include a simplified navigation mode and a user interface built around clarity, limiting sensory and cognitive overload.

The greatest ethical challenge, however, may arise from AI's own success. Features such as Vision AI learn, decide, and adapt to users; they may inevitably gather sensitive information about the conditions and behavior of people in vulnerable circumstances. Truly responsible innovation requires transparency: How is this information protected? Is it used for model training in ways that reproduce bias?

Another question is whether access to the best accessibility features—such as those found in premium Neo QLED 4K models—creates a new digital divide within disabled communities between people who can afford premium technology and those who cannot.

The impact of inclusive design becomes tangible in individual lives. One inspiring example is Sheridan O'Donnell, a director and writer diagnosed in 2021 with retinitis pigmentosa, a rare condition causing progressive vision loss [24]. Sheridan has explained how Voice Guide, Audio Description, and especially Relumino Mode enabled him to review his short film Flush and continue directing [25]. His experience illustrates that well-designed technology can do more than solve a functional problem: it can help preserve people's passions and professional identities.

It is at this intersection of human need and innovation that Samsung's “Tech for All” initiative seeks to provide accessible and inclusive technology through product design, features, and socially impactful programs [26]. Initiatives such as Vision AI suggest an encouraging future, even though considerable work remains, especially for people with intersecting disabilities. There are concrete steps to take: co-design with users with sensory, motor, and cognitive disabilities; publish more public measures of adoption, satisfaction, and autonomy; and communicate accessibility features so that people can discover and use them.

The most powerful innovation is not measured by hardware specifications or processing power, but by how many lives it enables people to see, hear, understand, and make decisions without barriers.

The 2.2 billion people with visual impairment are not an abstract statistic. They include people such as Sheridan O'Donnell and the urgent needs of deafblind communities. They remind us why innovation guided by AI and deeply human ethics is redefining what it means to see, hear, and participate in the world. That is the true vision of tomorrow.

References

The numbered references [1]–[26] refer to the bibliography in the official original Spanish essay. For exact bibliographic entries, consult: https://vinculacion.iimas.unam.mx/?p=5052